

Bitcrushers Milestone 1

<https://github.com/CSC415-2025-Fall/csc415-filesystem-shorbs>

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What's in the VCB

The VCB is a key component in our file system. We store it in block 0. Its job is to tell the system how the disk is laid out. When our file system boots up, the first thing it does is read this block to check the magic number. If the number matches our magic number (0x415F5), then it knows the disk is formatted and ready to go. Otherwise, it knows its blank and needs to run the formatting code.

What's in our VCB:

- magicNumber: This is our unique signature 0x415F5.
- blockSize: Size of a block.
- totalBlocks: Total blocks in volume.
- FATStart: Start block in FAT.
- FATBlocks: Size of FAT in blocks.
- dataStart: First block where data starts.
- rootStart: First block in root.
- rootSize: Size in bytes.
- freeBlocks: Free blocks.

Free Space Structure

For our free space structure we are using a FAT table to manage freespace. As of right now we have basic implementation of it but we plan to improve and optimize it. We have an allocateChain function that is our main method of getting free space. It allocates indexes to the first open spot and adds it to a chain. The memory is then written to the copy of FAT in ram and on disc. It then returns the head so files can use that chain.

Directory System

Our directory entry struct, `DirEntry`, includes flags, type, startBlock, size in blocks, size in bytes, name, nameLength, and an isDir flag. The function that I made for this milestone is `DirEntry *createDir` which takes in the minimum amount of entries and a pointer to the directory's parent. For this milestone we needed to create the root directory which is done in the `fsinit` function. We pass in 20 and NULL to the function. By passing in a null pointer as the parent we are representing the root not having a parent / it being its own parent.

To create a directory we need to allocate an array of directory entries. In order to allocate the correct amount of space we need to calculate the bytes to malloc and then optimize the amount of entries to take up as much of the block as possible. Then we set each of the names to '\0'. This represents unused directories. Each directory will have the "." and ".." entries representing itself and its parent respectively. If the parent pointer that we have passed in is NULL then we set the parent directory index to itself, or the one we are creating. Then we populate the second index ".." with the information provided from the parent pointer. While populating the fields of the new directory we consult the freespace to allocate a chain of blocks for us to use and the starting block number. Finally, we use `LBAwrite` to write to disk, the buffer is our array of entries, the size is provided by the calculations we made earlier and the location is given to us by the free space allocation (FAT).

Table

Change #	Name	Change made	 Date
1	Yani	Added VCB and DirEntry structs, defined VCB_MAGIC_NUMBER, defined function prototypes createBitmap, findFreeBlocks, and createRootDirectory	10/25/2025
2	Daniel	Started implementation of free space structure. Created header file for it and created function prototypes. Also got it to write and init the space on disk dedicated to the free space map.	10/26/2025
3	Yani	Started working on PDF, wrote about VCB structure	11/1/2025
4	Benji	Using the Code from class in conjunction with our own directory structure, implemented the createDir function that will write the directory to disk and handles if it is a root directory as well	11/2/2025
5	Daniel	Finished basic implmentation of the free space structure. Changed from bitmap to FAT. Added a function to allocate chains from the FAT.	11/3/2025
6	Shadi	Wrote Groupwork section of PDF, organized tables and proof read the final PDF submission.	11/3/2025

How our Team Handled Group Work

During our first meeting, we decided to split the work like this:

- Yani: In charge of setting up the mfs.h file with the VCB and DirEntry structs, including function prototypes, and defining the magic number.
- Daniel: In charge of all the FAT logic, also helped with the hex dump
- Benji: In charge of writing the createRootDirectory function and running the hex dump.
- Shadi: In charge of updating tables and organizing meetings, also working on/proofreading PDF.

In regards to workflow, our work revolved around the code in mfs.h, so once Yani pushed her code, the rest of the team was able to start working on their assigned tasks. We communicated frequently through our Discord server, where we had notes, questions, and GitHub updates all being shared, so everyone was on the same page. We created a table tracking our meetings:

Column 1	Column 2	Column 3	Column 4	🕒 Column 5	# Column 6	Column 7	Column 8
Meeting #	Date Met	Online/In-person	Location/Platform	When (Time)	How Long (Minutes)	Attendees	Purpose
1	10/23/25	In-person	Library	2:00:00 PM	120	Daniel, Yani, Benji	Plan, understand assignment more
2	10/27/25	Online	Discord	5:00:00 PM	30	Daniel, Yani, Benji, shady	Check in
3	11/3/25	Online	Discord	5:00:00 PM	120	Daniel, Yani, Benji	Code Review

Issues and resolutions

- Our first issue we faced is to fully understand the assignment and how all the moving parts fit together. One of the most confusing concepts we faced was figuring out what was on disc and what was on memory. After a long discussion we resolved a lot of our issues and questions by going over material and fully taking our time to understand it.
- When working on the free space map, we overthought our solution which ended up taking a lot of time. Our solution to this was to lower the scope a bit and implement a basic working solution and optimize it later.

Dump

VCB

[illegible]

FAT

000400:	01	02	03	04	05	FF	07	08	09	FF	00	00	00	00	00	00	00	*****?***?*****
000410:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000420:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000430:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000440:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000450:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000460:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000470:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000480:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000490:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
0004A0:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
0004B0:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
0004C0:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
0004D0:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
0004E0:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
0004F0:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000500:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000510:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000520:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000530:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000540:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000550:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000560:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000570:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000580:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
000590:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
0005A0:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
0005B0:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
0005C0:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
0005D0:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
0005E0:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****
0005F0:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	*****

The chains in the FAT are values separated by FF

[illegible]

[illegible]

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002400: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
002410: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
002420: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
002430: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
002440: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
002450: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
002460: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
002470: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
002480: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
002490: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0024A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0024B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0024C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0024D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0024E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0024F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
```


002800:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
002810:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
002820:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
002830:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
002840:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
002850:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
002860:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
002870:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
002880:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
002890:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
0028A0:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
0028B0:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
0028C0:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
0028D0:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
0028E0:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****
0028F0:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	*****

[illegible][illegible][illegible][illegible]

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003800: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
003810: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
003820: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
003830: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
003840: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
003850: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
003860: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
003870: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
003880: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
003890: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0038A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0038B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0038C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0038D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0038E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0038F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]


```
006800: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
006810: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
006820: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
006830: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
006840: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
006850: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
006860: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
006870: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
006880: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
006890: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0068A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0068B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0068C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0068D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0068E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0068F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 78 for 1 block:

009C00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009C10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009C20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009C30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009C40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009C50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009C60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009C70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009C80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009C90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009CA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009CB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009CC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009CD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009CE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009CF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

009D00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009D10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009D20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009D30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009D40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009D50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009D60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009D70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009D80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009D90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009DA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009DB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009DC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009DD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009DE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009DF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 79 for 1 block:

009E00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009E10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009E20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009E30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009E40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009E50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009E60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009E70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009E80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009E90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009EA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009EB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009EC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009ED0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009EE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009EF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

009F00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009F10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009F20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009F30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009F40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009F50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009F60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009F70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009F80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009F90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009FA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009FB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009FC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009FD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009FE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
009FF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```



```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 80 for 1 block:

00A000: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A010: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A050: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A060: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A070: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A080: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A090: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A0A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A0B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A0C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A0D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A0E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A0F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00A100: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A120: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A130: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A140: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A150: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A170: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A180: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A190: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A1A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A1B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A1C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A1D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A1E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A1F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 81 for 1 block:

00A200: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A210: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A220: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A230: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A240: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A250: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A260: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A270: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A280: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A290: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A2A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A2B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A2C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A2D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A2E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A2F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00A300: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A310: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A320: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A330: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A340: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A350: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A360: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A370: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A380: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A390: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A3A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A3B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A3C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A3D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A3E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A3F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 82 for 1 block:

00A400: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A410: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A420: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A430: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A440: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A450: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A460: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A470: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A480: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A490: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A4A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A4B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A4C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A4D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A4E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A4F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00A500: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A510: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A520: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A530: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A540: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A550: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A560: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A570: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A580: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A590: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A5A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A5B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A5C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A5D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A5E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A5F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 83 for 1 block:

00A600: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A610: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A620: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A630: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A640: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A650: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A660: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A670: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A680: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A690: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A6A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A6B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A6C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A6D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A6E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A6F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00A700: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A710: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A720: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A730: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A740: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A750: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A760: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A770: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A780: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A790: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A7A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A7B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A7C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A7D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A7E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A7F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 84 for 1 block:

00A800: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A810: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A820: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A830: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A840: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A850: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A860: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A870: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A880: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A890: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A8A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A8B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A8C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A8D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A8E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A8F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00A900: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A910: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A920: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A930: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A940: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A950: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A960: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A970: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A980: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A990: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A9A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A9B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A9C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A9D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A9E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00A9F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 85 for 1 block:

00AA00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AA10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AA20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AA30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AA40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AA50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AA60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AA70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AA80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AA90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AAA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AAB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AAC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AAD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AAE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AAF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00AB00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AB10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AB20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AB30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AB40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AB50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AB60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AB70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AB80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AB90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ABA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ABB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ABC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ABD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ABE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ABF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 86 for 1 block:

00AC00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AC10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AC20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AC30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AC40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AC50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AC60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AC70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AC80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AC90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ACA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ACB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ACC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ACD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ACE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ACF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00AD00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AD10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AD20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AD30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AD40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AD50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AD60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AD70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AD80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AD90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ADA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ADB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ADC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ADD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ADE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ADF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 87 for 1 block:

00AE00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AE10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AE20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AE30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AE40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AE50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AE60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AE70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AE80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AE90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AEA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AEB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AEC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AED0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AEE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AEF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00AF00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AF10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AF20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AF30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AF40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AF50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AF60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AF70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AF80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AF90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AFA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AFB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AFC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AFD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AFE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00AFF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```



```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 88 for 1 block:

00B000: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B010: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B050: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B060: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B070: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B080: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B090: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B0A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B0B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B0C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B0D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B0E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B0F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00B100: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B120: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B130: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B140: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B150: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B170: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B180: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B190: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B1A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B1B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B1C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B1D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B1E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B1F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 89 for 1 block:

00B200: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B210: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B220: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B230: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B240: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B250: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B260: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B270: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B280: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B290: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B2A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B2B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B2C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B2D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B2E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B2F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00B300: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B310: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B320: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B330: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B340: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B350: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B360: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B370: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B380: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B390: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B3A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B3B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B3C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B3D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B3E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B3F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 90 for 1 block:

00B400: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B410: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B420: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B430: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B440: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B450: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B460: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B470: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B480: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B490: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B4A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B4B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B4C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B4D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B4E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B4F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00B500: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B510: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B520: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B530: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B540: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B550: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B560: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B570: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B580: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B590: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B5A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B5B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B5C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B5D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B5E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B5F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 91 for 1 block:

00B600: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B610: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B620: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B630: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B640: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B650: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B660: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B670: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B680: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B690: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B6A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B6B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B6C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B6D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B6E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B6F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00B700: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B710: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B720: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B730: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B740: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B750: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B760: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B770: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B780: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B790: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B7A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B7B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B7C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B7D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B7E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B7F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 92 for 1 block:

00B800: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B810: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B820: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B830: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B840: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B850: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B860: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B870: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B880: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B890: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B8A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B8B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B8C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B8D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B8E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B8F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00B900: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B910: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B920: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B930: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B940: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B950: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B960: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B970: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B980: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B990: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B9A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B9B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B9C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B9D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B9E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00B9F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 93 for 1 block:

00BA00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BA10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BA20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BA30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BA40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BA50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BA60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BA70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BA80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BA90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BAA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BAB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BAC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BAD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BAE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BAF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00BB00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BB10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BB20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BB30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BB40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BB50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BB60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BB70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BB80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BB90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BBA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BBB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BBC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BBD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BBE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BBF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 94 for 1 block:

00BC00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BC10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BC20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BC30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BC40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BC50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BC60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BC70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BC80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BC90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BCA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BCB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BCC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BCD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BCE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BCF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00BD00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BD10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BD20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BD30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BD40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BD50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BD60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BD70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BD80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BD90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BDA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BDB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BDC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BDD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BDE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BDF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 95 for 1 block:

00BE00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BE10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BE20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BE30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BE40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BE50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BE60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BE70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BE80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BE90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BEA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BEB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BEC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BED0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BEE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BEF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00BF00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BF10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BF20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BF30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BF40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BF50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BF60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BF70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BF80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BF90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BFA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BFB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BFC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BFD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BFE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BFF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```




student@student: ~/Desktop/CSC...



student@student:~/Desktop/CSC415/CSC415-Filesystem-shorbs\$./hexdump/hexdumpL

SampleVolume --start 96 --count 1

Dumping file SampleVolume, starting at block 96 for 1 block:

```
00C000: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C010: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C050: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C060: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C070: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C080: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C090: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C0A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C0B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C0C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C0D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C0E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C0F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00C100: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C120: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C130: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C140: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C150: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C170: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C180: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C190: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C1A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C1B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C1C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C1D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C1E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C1F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 97 for 1 block:

00C200: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C210: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C220: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C230: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C240: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C250: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C260: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C270: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C280: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C290: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C2A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C2B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C2C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C2D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C2E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C2F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00C300: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C310: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C320: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C330: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C340: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C350: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C360: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C370: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C380: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C390: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C3A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C3B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C3C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C3D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C3E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C3F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 98 for 1 block:

00C400: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C410: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C420: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C430: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C440: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C450: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C460: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C470: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C480: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C490: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C4A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C4B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C4C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C4D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C4E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C4F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00C500: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C510: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C520: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C530: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C540: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C550: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C560: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C570: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C580: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C590: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C5A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C5B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C5C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C5D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C5E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C5F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 99 for 1 block:

00C600: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C610: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C620: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C630: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C640: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C650: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C660: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C670: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C680: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C690: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C6A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C6B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C6C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C6D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C6E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C6F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00C700: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C710: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C720: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C730: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C740: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C750: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C760: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C770: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C780: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C790: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C7A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C7B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C7C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C7D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C7E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C7F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 100 for 1 block:

00C800: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C810: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C820: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C830: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C840: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C850: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C860: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C870: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C880: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C890: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C8A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C8B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C8C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C8D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C8E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C8F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00C900: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C910: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C920: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C930: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C940: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C950: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C960: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C970: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C980: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C990: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C9A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C9B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C9C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C9D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C9E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00C9F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 101 for 1 block:

00CA00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CA10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CA20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CA30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CA40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CA50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CA60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CA70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CA80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CA90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CAA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CAB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CAC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CAD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CAE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CAF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00CB00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CB10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CB20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CB30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CB40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CB50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CB60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CB70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CB80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CB90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CBA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CBB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CBC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CBD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CBE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CBF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 102 for 1 block:

00CC00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CC10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CC20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CC30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CC40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CC50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CC60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CC70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CC80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CC90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CCA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CCB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CCC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CCD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CCE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CCF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00CD00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CD10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CD20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CD30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CD40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CD50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CD60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CD70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CD80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CD90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CDA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CDB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CDC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CDD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CDE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CDF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 103 for 1 block:

00CE00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CE10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CE20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CE30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CE40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CE50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CE60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CE70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CE80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CE90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CEA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CEB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CEC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CED0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CEE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CEF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00CF00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CF10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CF20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CF30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CF40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CF50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CF60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CF70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CF80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CF90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CFA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CFB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CFC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CFD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CFE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00CFF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```



```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 104 for 1 block:

00D000: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D010: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D050: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D060: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D070: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D080: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D090: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D0A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D0B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D0C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D0D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D0E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D0F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00D100: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D120: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D130: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D140: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D150: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D170: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D180: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D190: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D1A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D1B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D1C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D1D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D1E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D1F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 105 for 1 block:

00D200: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D210: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D220: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D230: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D240: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D250: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D260: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D270: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D280: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D290: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D2A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D2B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D2C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D2D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D2E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D2F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00D300: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D310: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D320: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D330: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D340: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D350: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D360: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D370: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D380: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D390: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D3A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D3B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D3C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D3D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D3E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D3F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 106 for 1 block:

00D400: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D410: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D420: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D430: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D440: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D450: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D460: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D470: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D480: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D490: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D4A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D4B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D4C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D4D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D4E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D4F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00D500: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D510: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D520: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D530: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D540: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D550: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D560: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D570: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D580: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D590: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D5A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D5B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D5C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D5D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D5E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D5F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 107 for 1 block:

00D600: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D610: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D620: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D630: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D640: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D650: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D660: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D670: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D680: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D690: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D6A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D6B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D6C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D6D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D6E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D6F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00D700: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D710: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D720: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D730: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D740: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D750: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D760: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D770: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D780: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D790: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D7A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D7B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D7C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D7D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D7E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D7F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 108 for 1 block:

00D800: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D810: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D820: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D830: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D840: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D850: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D860: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D870: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D880: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D890: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D8A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D8B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D8C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D8D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D8E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D8F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00D900: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D910: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D920: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D930: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D940: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D950: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D960: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D970: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D980: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D990: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D9A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D9B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D9C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D9D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D9E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00D9F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 109 for 1 block:

00DA00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DA10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DA20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DA30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DA40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DA50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DA60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DA70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DA80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DA90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DAA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DAB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DAC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DAD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DAE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DAF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00DB00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DB10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DB20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DB30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DB40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DB50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DB60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DB70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DB80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DB90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DBA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DBB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DBC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DBD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DBE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DBF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 110 for 1 block:

00DC00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DC10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DC20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DC30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DC40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DC50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DC60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DC70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DC80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DC90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DCA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DCB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DCC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DCD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DCE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DCF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00DD00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DD10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DD20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DD30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DD40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DD50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DD60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DD70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DD80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DD90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DDA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ddb0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DDC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DDD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DDE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DDF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 111 for 1 block:

00DE00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DE10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DE20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DE30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DE40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DE50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DE60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DE70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DE80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DE90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DEA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DEB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DEC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DED0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DEE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DEF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00DF00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DF10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DF20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DF30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DF40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DF50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DF60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DF70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DF80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DF90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DFA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DFB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DFC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DFD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DFE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00DFF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```



```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 112 for 1 block:

00E000: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E010: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E050: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E060: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E070: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E080: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E090: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E0A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E0B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E0C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E0D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E0E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E0F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00E100: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E120: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E130: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E140: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E150: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E170: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E180: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E190: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E1A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E1B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E1C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E1D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E1E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E1F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 113 for 1 block:

00E200: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E210: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E220: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E230: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E240: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E250: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E260: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E270: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E280: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E290: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E2A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E2B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E2C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E2D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E2E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E2F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00E300: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E310: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E320: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E330: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E340: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E350: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E360: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E370: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E380: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E390: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E3A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E3B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E3C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E3D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E3E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E3F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 114 for 1 block:

00E400: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E410: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E420: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E430: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E440: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E450: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E460: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E470: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E480: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E490: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E4A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E4B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E4C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E4D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E4E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E4F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00E500: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E510: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E520: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E530: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E540: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E550: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E560: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E570: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E580: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E590: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E5A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E5B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E5C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E5D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E5E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E5F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 115 for 1 block:

00E600: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E610: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E620: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E630: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E640: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E650: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E660: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E670: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E680: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E690: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E6A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E6B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E6C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E6D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E6E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E6F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00E700: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E710: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E720: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E730: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E740: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E750: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E760: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E770: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E780: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E790: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E7A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E7B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E7C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E7D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E7E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E7F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 116 for 1 block:

00E800: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E810: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E820: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E830: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E840: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E850: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E860: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E870: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E880: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E890: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E8A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E8B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E8C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E8D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E8E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E8F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00E900: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E910: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E920: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E930: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E940: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E950: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E960: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E970: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E980: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E990: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E9A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E9B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E9C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E9D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E9E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00E9F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 117 for 1 block:

00EA00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EA10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EA20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EA30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EA40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EA50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EA60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EA70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EA80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EA90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EAA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EAB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EAC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EAD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EAE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EAF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00EB00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EB10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EB20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EB30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EB40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EB50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EB60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EB70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EB80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EB90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EBA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EBB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EBC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EBD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EBE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EBF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 118 for 1 block:

00EC00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EC10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EC20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EC30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EC40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EC50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EC60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EC70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EC80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EC90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ECA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ECB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ECC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ECD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ECE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ECF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00ED00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ED10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ED20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ED30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ED40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ED50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ED60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ED70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ED80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00ED90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EDA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EDB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EDC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EDD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EDE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EDF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 119 for 1 block:

00EE00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EE10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EE20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EE30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EE40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EE50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EE60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EE70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EE80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EE90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EEA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EEB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EEC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EED0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EEE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EEF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00EF00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EF10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EF20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EF30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EF40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EF50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EF60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EF70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EF80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EF90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EFA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EFB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EFC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EFD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EFE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00EFF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```



```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 120 for 1 block:

00F000: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F010: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F050: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F060: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F070: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F080: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F090: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F0A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F0B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F0C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F0D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F0E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F0F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00F100: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F120: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F130: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F140: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F150: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F170: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F180: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F190: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F1A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F1B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F1C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F1D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F1E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F1F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 121 for 1 block:

00F200: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F210: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F220: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F230: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F240: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F250: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F260: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F270: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F280: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F290: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F2A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F2B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F2C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F2D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F2E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F2F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00F300: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F310: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F320: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F330: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F340: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F350: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F360: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F370: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F380: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F390: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F3A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F3B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F3C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F3D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F3E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F3F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 122 for 1 block:

00F400: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F410: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F420: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F430: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F440: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F450: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F460: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F470: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F480: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F490: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F4A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F4B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F4C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F4D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F4E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F4F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00F500: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F510: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F520: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F530: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F540: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F550: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F560: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F570: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F580: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F590: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F5A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F5B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F5C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F5D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F5E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F5F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 123 for 1 block:

00F600: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F610: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F620: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F630: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F640: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F650: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F660: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F670: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F680: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F690: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F6A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F6B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F6C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F6D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F6E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F6F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00F700: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F710: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F720: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F730: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F740: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F750: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F760: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F770: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F780: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F790: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F7A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F7B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F7C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F7D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F7E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F7F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 124 for 1 block:

00F800: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F810: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F820: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F830: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F840: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F850: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F860: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F870: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F880: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F890: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F8A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F8B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F8C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F8D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F8E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F8F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00F900: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F910: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F920: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F930: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F940: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F950: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F960: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F970: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F980: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F990: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F9A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F9B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F9C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F9D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F9E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00F9F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 125 for 1 block:

00FA00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FA10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FA20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FA30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FA40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FA50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FA60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FA70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FA80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FA90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FAA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FAB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FAC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FAD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FAE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FAF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00FB00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FB10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FB20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FB30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FB40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FB50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FB60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FB70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FB80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FB90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FBA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FBB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FBC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FBD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FBE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00BF00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 126 for 1 block:

00FC00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FC10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FC20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FC30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FC40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FC50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FC60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FC70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FC80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FC90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FCA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FCB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FCC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FCD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FCE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FCF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00FD00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FD10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FD20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FD30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FD40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FD50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FD60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FD70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FD80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FD90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FDA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FDB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FDC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FDD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FDE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FDF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 127 for 1 block:

00FE00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FE10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FE20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FE30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FE40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FE50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FE60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FE70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FE80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FE90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FEA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FEB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FEC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FED0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FEE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FEF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

00FF00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FF10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FF20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FF30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FF40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FF50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FF60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FF70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FF80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FF90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FFA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FFB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FFC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FFD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FFE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
00FFF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```



```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 128 for 1 block:

010000: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010010: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010050: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010060: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010070: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010080: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010090: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0100A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0100B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0100C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0100D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0100E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0100F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

010100: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010120: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010130: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010140: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010150: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010170: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010180: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010190: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0101A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0101B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0101C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0101D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0101E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0101F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 129 for 1 block:

010200: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010210: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010220: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010230: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010240: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010250: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010260: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010270: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010280: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010290: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0102A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0102B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0102C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0102D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0102E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0102F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

010300: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010310: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010320: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010330: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010340: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010350: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010360: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010370: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010380: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010390: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0103A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0103B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0103C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0103D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0103E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0103F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 130 for 1 block:

010400: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010410: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010420: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010430: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010440: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010450: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010460: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010470: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010480: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010490: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0104A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0104B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0104C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0104D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0104E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0104F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

010500: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010510: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010520: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010530: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010540: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010550: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010560: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010570: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010580: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010590: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0105A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0105B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0105C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0105D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0105E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0105F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 131 for 1 block:

010600: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010610: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010620: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010630: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010640: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010650: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010660: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010670: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010680: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010690: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0106A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0106B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0106C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0106D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0106E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0106F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

010700: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010710: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010720: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010730: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010740: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010750: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010760: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010770: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010780: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010790: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0107A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0107B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0107C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0107D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0107E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0107F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 132 for 1 block:

010800: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010810: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010820: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010830: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010840: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010850: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010860: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010870: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010880: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010890: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0108A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0108B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0108C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0108D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0108E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0108F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

010900: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010910: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010920: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010930: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010940: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010950: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010960: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010970: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010980: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010990: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0109A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0109B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0109C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0109D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0109E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0109F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 133 for 1 block:

010A00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010A10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010A20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010A30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010A40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010A50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010A60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010A70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010A80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010A90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010AA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010AB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010AC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010AD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010AE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010AF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

010B00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010B10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010B20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010B30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010B40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010B50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010B60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010B70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010B80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010B90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010BA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010BB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010BC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010BD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010BE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010BF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 134 for 1 block:

010C00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010C10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010C20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010C30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010C40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010C50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010C60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010C70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010C80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010C90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010CA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010CB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010CC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010CD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010CE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010CF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

010D00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010D10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010D20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010D30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010D40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010D50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010D60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010D70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010D80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010D90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010DA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010DB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010DC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010DD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010DE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010DF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 135 for 1 block:

010E00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010E10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010E20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010E30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010E40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010E50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010E60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010E70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010E80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010E90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010EA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010EB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010EC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010ED0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010EE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010EF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

010F00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010F10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010F20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010F30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010F40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010F50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010F60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010F70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010F80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010F90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010FA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010FB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010FC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010FD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010FE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
010FF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```



```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 136 for 1 block:

011000: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011010: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011050: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011060: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011070: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011080: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011090: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0110A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0110B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0110C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0110D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0110E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0110F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

011100: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011120: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011130: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011140: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011150: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011170: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011180: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011190: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0111A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0111B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0111C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0111D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0111E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0111F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 137 for 1 block:

011200: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011210: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011220: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011230: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011240: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011250: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011260: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011270: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011280: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011290: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0112A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0112B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0112C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0112D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0112E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0112F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

011300: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011310: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011320: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011330: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011340: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011350: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011360: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011370: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011380: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011390: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0113A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0113B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0113C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0113D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0113E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0113F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 138 for 1 block:

011400: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011410: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011420: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011430: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011440: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011450: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011460: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011470: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011480: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011490: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0114A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0114B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0114C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0114D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0114E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0114F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

011500: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011510: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011520: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011530: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011540: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011550: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011560: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011570: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011580: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011590: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0115A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0115B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0115C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0115D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0115E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0115F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 139 for 1 block:

011600: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011610: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011620: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011630: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011640: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011650: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011660: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011670: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011680: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011690: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0116A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0116B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0116C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0116D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0116E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0116F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

011700: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011710: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011720: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011730: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011740: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011750: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011760: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011770: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011780: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011790: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0117A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0117B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0117C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0117D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0117E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0117F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 140 for 1 block:

011800: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011810: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011820: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011830: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011840: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011850: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011860: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011870: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011880: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011890: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0118A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0118B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0118C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0118D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0118E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0118F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

011900: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011910: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011920: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011930: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011940: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011950: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011960: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011970: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011980: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011990: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0119A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0119B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0119C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0119D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0119E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0119F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 141 for 1 block:

011A00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011A10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011A20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011A30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011A40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011A50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011A60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011A70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011A80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011A90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011AA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011AB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011AC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011AD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011AE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011AF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

011B00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011B10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011B20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011B30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011B40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011B50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011B60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011B70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011B80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011B90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011BA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011BB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011BC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011BD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011BE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011BF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 142 for 1 block:

011C00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011C10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011C20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011C30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011C40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011C50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011C60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011C70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011C80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011C90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011CA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011CB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011CC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011CD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011CE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011CF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

011D00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011D10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011D20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011D30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011D40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011D50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011D60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011D70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011D80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011D90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011DA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011DB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011DC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011DD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011DE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011DF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 143 for 1 block:

011E00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011E10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011E20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011E30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011E40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011E50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011E60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011E70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011E80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011E90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011EA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011EB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011EC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011ED0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011EE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011EF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

011F00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011F10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011F20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011F30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011F40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011F50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011F60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011F70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011F80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011F90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011FA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011FB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011FC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011FD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011FE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
011FF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```



```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 144 for 1 block:

012000: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012010: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012050: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012060: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012070: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012080: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012090: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0120A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0120B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0120C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0120D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0120E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0120F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

012100: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012120: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012130: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012140: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012150: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012170: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012180: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012190: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0121A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0121B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0121C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0121D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0121E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0121F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 145 for 1 block:

012200: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012210: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012220: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012230: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012240: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012250: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012260: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012270: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012280: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012290: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0122A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0122B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0122C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0122D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0122E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0122F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

012300: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012310: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012320: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012330: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012340: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012350: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012360: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012370: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012380: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012390: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0123A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0123B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0123C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0123D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0123E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0123F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 146 for 1 block:

012400: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012410: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012420: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012430: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012440: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012450: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012460: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012470: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012480: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012490: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0124A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0124B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0124C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0124D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0124E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0124F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

012500: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012510: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012520: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012530: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012540: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012550: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012560: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012570: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012580: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012590: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0125A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0125B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0125C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0125D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0125E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0125F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 147 for 1 block:

012600: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012610: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012620: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012630: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012640: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012650: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012660: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012670: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012680: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012690: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0126A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0126B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0126C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0126D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0126E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0126F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

012700: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012710: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012720: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012730: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012740: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012750: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012760: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012770: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012780: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012790: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0127A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0127B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0127C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0127D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0127E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0127F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 148 for 1 block:

012800: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012810: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012820: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012830: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012840: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012850: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012860: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012870: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012880: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012890: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0128A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0128B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0128C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0128D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0128E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0128F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

012900: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012910: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012920: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012930: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012940: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012950: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012960: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012970: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012980: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012990: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0129A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0129B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0129C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0129D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0129E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0129F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 149 for 1 block:

012A00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012A10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012A20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012A30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012A40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012A50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012A60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012A70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012A80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012A90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012AA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012AB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012AC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012AD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012AE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012AF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

012B00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012B10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012B20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012B30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012B40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012B50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012B60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012B70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012B80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012B90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012BA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012BB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012BC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012BD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012BE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012BF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 150 for 1 block:

012C00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012C10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012C20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012C30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012C40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012C50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012C60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012C70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012C80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012C90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012CA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012CB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012CC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012CD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012CE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012CF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

012D00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012D10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012D20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012D30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012D40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012D50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012D60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012D70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012D80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012D90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012DA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012DB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012DC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012DD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012DE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012DF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 151 for 1 block:

012E00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012E10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012E20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012E30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012E40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012E50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012E60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012E70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012E80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012E90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012EA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012EB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012EC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012ED0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012EE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012EF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

012F00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012F10: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012F20: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012F30: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012F40: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012F50: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012F60: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012F70: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012F80: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012F90: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012FA0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012FB0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012FC0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012FD0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012FE0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
012FF0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```



```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 152 for 1 block:

013000: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013010: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013050: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013060: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013070: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013080: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013090: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0130A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0130B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0130C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0130D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0130E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0130F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

013100: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013120: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013130: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013140: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013150: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013170: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013180: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013190: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0131A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0131B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0131C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0131D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0131E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0131F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 153 for 1 block:

013200: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013210: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013220: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013230: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013240: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013250: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013260: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013270: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013280: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013290: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0132A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0132B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0132C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0132D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0132E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0132F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

013300: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013310: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013320: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013330: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013340: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013350: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013360: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013370: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013380: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013390: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0133A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0133B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0133C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0133D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0133E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0133F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 154 for 1 block:

013400: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013410: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013420: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013430: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013440: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013450: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013460: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013470: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013480: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013490: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0134A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0134B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0134C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0134D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0134E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0134F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

013500: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013510: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013520: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013530: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013540: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013550: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013560: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013570: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013580: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013590: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0135A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0135B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0135C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0135D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0135E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0135F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

```
student@student: ~/Desktop/CSC...
Dumping file SampleVolume, starting at block 155 for 1 block:

013600: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013610: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013620: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013630: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013640: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013650: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013660: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013670: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013680: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013690: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0136A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0136B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0136C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0136D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0136E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0136F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

013700: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013710: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013720: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013730: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013740: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013750: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013760: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013770: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013780: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
013790: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0137A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0137B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0137C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0137D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0137E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0137F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$
```

Root Directory

```

student@student:~/Desktop/CSC415/csc415-filesystem-shorbs$ ./Hexdump/hexdump1 SampleVolume --start 161 --count 3
Dumping file SampleVolume, starting at block 161 for 3 blocks:

014200: 00 00 00 00 00 00 00 00 06 00 00 00 D8 05 00 00 | *****?***
014210: 03 00 00 00 2E 00 00 00 00 00 00 00 00 00 00 00 | ****.*****
014220: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
014230: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
014240: 01 00 00 00 00 00 00 00 00 00 00 00 06 00 00 00 | *****
014250: D8 05 00 00 03 00 00 00 2E 2E 00 00 00 00 00 00 | ?*****.*****
014260: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
014270: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
014280: 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 | *****
014290: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0142A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0142B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0142C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0142D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0142E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****
0142F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 | *****

```

As expected we can see the root directory is in block 161 after the VCB and FAT and the first two directory entries in that block are “.” and “..”. We know that the position of these names is correct because it is 20 bytes (5 ints) from the start. We can also see the size in blocks (3) that comes before the name nibbles. Looking further than that for both entries we see that there is an 01 representing the isDir field that is set to 1. On 14210 there is an 03 that represents that the directory takes up 03 blocks in total.